ARYAN BHAVSAR

AI/ML Engineer | Forward-Deployed AI | AI Solutions Engineer
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LinkedIn | GitHub | LeetCode | Portfolio

EDUCATION

Vellore Institute of Technology, Chennai **2021 – 2025**
B.Tech in Computer Science and Engineering CGPA: **8.65**

PUBLICATIONS

- Deep Learning for Tea Leaf Disease Synthesis & Detection – ScienceDirect
- Markerless Data-based Cyclist Posture Detection using Deep Learning – IEEE

PROFESSIONAL EXPERIENCE

Hitech Digital Solutions **Mar 2026 – Present**
Machine Learning Systems Engineer – Forward-Deployed AI & Full Stack *Ahmedabad, India*

- Architected and built a production-grade Real Estate Document Intelligence Platform as the primary ML/full-stack engineer in a 4-member team, owning React/Vite frontend, FastAPI backend, SQLite job orchestration, ML pipelines, and deployment for deed and mortgage workflows.
- Replaced paid Azure OCR and legacy BERT classification with open-source document AI and vision-language models (DocTR, Donut, Monkey OCR, Qwen 2/2.5/3, Mistral, InternVL), achieving **90%+ extraction accuracy** while eliminating third-party inference API costs.
- Built model-serving pipelines using vLLM, LoRA fine-tuning with Unsloth, sidecar health monitoring, and zero-downtime hot-swapping; reduced deed processing from **15s to 1s**, mortgage from **25s to 6s**, and classification from **5s to 0.5s**.
- Enabled **50,000+ documents/day** in production and implemented confidence-based manual review routing, contributing to verification headcount reduction from **500 to under 100**, a **33% reduction in operational costs**, and a **10% increase in profit**.
- Led technical discovery and requirements discussions with stakeholders and clients; translated needs into BRDs, solution architecture, and technical documentation; conducted demos and UAT, coordinated feedback, and supported Docker/RunPod deployment and production operations. Awarded **Star Performer of the Month and Quarter**.

Hitech Digital Solutions **Sep 2025 – Feb 2026**
Graduate Engineer Trainee – Data Science *Ahmedabad, India*

- Developed an intelligent pricing and changelog management system, benchmarking open-source models and implementing GPT-4o Vision extraction, Sentence-Transformer semantic matching, and hybrid reconciliation with Pydantic schemas to automate menu data entry and track price changes.

Hyperface Technologies Pvt. Ltd. **Feb 2025 – Aug 2025**
AI Researcher – Generative AI Team *Bengaluru, India*

- Contributed to HyperAgent, a FinTech voice-AI platform supporting digital bank onboarding, Cards-as-a-Service (CaaS) journeys, card servicing, NLP-agent-driven cross-selling in place of traditional BPM workflows, and automated rewards management.
- Developed FastAPI-based multi-agent workflows and orchestration; implemented RAG/CAG patterns, prompt engineering, and context management for customer interactions and workflow execution.
- Built LLM evaluation and benchmarking workflows using Hugging Face models; explored caching and inference strategies to improve response quality and latency.

PROJECTS

BarneyWingman – Multimodal Dating Assistant (In Development) GitHub

- Building a human-in-the-loop, vision-powered pipeline that ingests multi-frame screenshots, extracts and merges profile evidence, and evaluates compatibility across dating platforms.
- Uses the consolidated profile context to generate personalised conversation starters, with the user retaining control over review and interaction.

Virtual Football Analysis GitHub

- Developed YOLOv8 and OpenCV-based player/ball tracking with a Flask application for match-event analysis and automated highlight generation.

TECHNICAL SKILLS

Programming: Python, Java, C++, SQL, JavaScript, HTML, CSS
Generative AI: RAG, CAG, Multi-Agent Systems, Prompt Engineering, Context Engineering, LLM Evaluation, Embeddings, LangChain
ML: PyTorch, TensorFlow, Scikit-learn, CNNs, NLP, Computer Vision, Sentence Transformers, Hugging Face, OpenCV
Inference & Deployment: vLLM, SGLang, LoRA, Model Serving, GPU Inference, Quantisation, Docker, RunPod, Linux
Backend & Data: FastAPI, Flask, React, REST APIs, PostgreSQL, MySQL, SQLite, Git, GitHub Actions (CI/CD), Vercel
Product & Delivery: Technical Discovery, Requirements Gathering, Solution Architecture, Client Demos, Stakeholder Engagement, UAT, Cross-functional Collaboration, Production Support